

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☐ | ☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☐ | ☒ A description of all covariates tested |
| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Scripts for simulations and example real-data analyses are available at https://github.com/anikamis/ChronoAdd/tree/main . Zenodo: DOI: 10.5281/zenodo.14681071
Data analysis	We used the following softwares for handling and scoring the genetic data: Plink 2.0, R 4.3.1 (including package PRSmix), and Python 3.11.5 (including packages pandas 2.1.4 and NumPy 1.6.2). For data analysis and visualization, we used R 4.3.1 (including packages PRSmix, ggplot2, meta, class, rcompanion, survminer) and Python 3.11.5 (including packages pandas 2.1.4, NumPy 1.6.2, and Matplotlib 3.7.2). Analyses were performed on the AoU Researcher Workbench. Results are reported in compliance with the AoU Data and Statistics Dissemination Policy.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

This research was approved by the Mass General Brigham institutional review board. All data are made available from the All of Us Research Study to researchers from universities and other institutions with genuine research inquiries following institutional review board and All of Us approval within the All of Us workbench database under <https://www.researchallofus.org/data-tools/workbench>. This study used srWGS data from the All of Us Controlled Tier Dataset version 7 release. The weights of the polygenic scores analyzed in this study are publicly available for download from the Polygenic Score Catalog. The mixing weights of the ChronoAdd(Pubn) are available in the Supplementary Data 1-3.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Our manuscript is compliant with the journal's policy on sex and gender reporting and sex was carefully considered in our study design. All of Us provides both sex assigned at birth and self reported gender. Sex assigned at birth was used for all relevant analyses, including only individuals where the self reported gender matched sex assigned at birth. Our models use sex assigned at birth as a covariate in both PRSMix training and later regression analysis. Among the total cohort (n = 236,393), 61% (n = 143,375) are assigned female at birth.

Reporting on race, ethnicity, or other socially relevant groupings

For population-specific analyses, our manuscript utilized inferred ancestry groupings based on genetic similarity. From The 1000 Genomes Project, 20 principal components were derived based on genetic similarity and then projected onto the All of Us data. Using these principal components as input, the knn method (from the R package "class" version 7.3) performed k-nearest neighbor classification in order to infer genetic ancestry in the All of Us data using 1000 Genomes data as a reference panel. Thus, the ancestry groups utilized in our analysis are those of the five 1000 Genomes super-populations: AFR, AMR, EUR, EAS, and SAS. Using these inferred groups, the total cohort (n = 236,393) is of 23% AFR (n = 54,232), 19% AMR (n = 44,791), 2% EAS (n = 5,483), 55% EUR (n = 129,370), and 1% SAS (n = 2,517).

Population characteristics

Of the total cohort (n=236,393; mean enrollment age = 51.8 years; 61% female), all PRSMix training was performed in the training cohort comprising 80% (n=189,114; mean age = 51.8 years; 61% female) of the participants. This training cohort is 23% AFR [n=43,393; mean age = 49.3; 58% female; (prevalence = 4.7% CAD; 11.3% T2DM; 11.7% MDD)], 19% AMR [n = 35,841; mean age = 44.7; 67% female; (prevalence = 3.0% CAD; 10.2% T2DM; 12.2% MDD)], 2% EAS [n = 4,330; mean age = 44.5; 64% female; (prevalence = 1.9% CAD; 5.9% T2DM; 5.2% MDD)], 55% EUR [n = 103,553; mean age = 55.8; 60% female; (prevalence = 5.7% CAD; 8.4% T2DM; 15.7% MDD)], and 1% SAS [n = 1,997; mean age = 42.5; 50% female; (prevalence = 3.1% CAD; 9.0% T2DM; 6.0% MDD)]. All results in the manuscript were calculated in the holdout test cohort comprising the remaining 20% (n = 47,279; mean age = 51.8; 61% female) of participants. This testing cohort is 23% AFR [n = 10,839; mean age = 49.2; 57% female; (prevalence = 4.8% CAD; 10.6% T2DM; 12.3% MDD)], 19% AMR [n = 8,950; mean age = 44.8; 67% female; (prevalence = 3.4% CAD; 9.7% T2DM; 11.7% MDD)], 2% EAS [n = 1,153; mean age = 44.3; 64% female; (prevalence = 1.6% CAD; 5.6% T2DM; 5.6% MDD)], 55% EUR [n = 25,817; mean age = 55.8; 60% female; (prevalence = 5.4% CAD; 8.2% T2DM; 15.6% MDD)], and 1% SAS [n = 520; mean age = 41.4; 50% female; (prevalence = 2.5% CAD; 6.3% T2DM; 5.2% MDD)]. All analyses were carried out in subgroups of inferred genetic ancestry and the first 10 principal components of genetic ancestry were used as covariates in regression analyses.

Recruitment

All of Us opened for enrollment in May 2018 and currently enrolls participants 18 years of age or older from a network of more than 730 recruitment sites. Elements of the program protocol include health questionnaires, electronic health records (EHRs), physical measurements, the use of digital health technology, and the collection and analysis of biospecimens. As of July 2024, more than 560,000 participants had contributed biospecimens. More than 80% of these participants are from groups that have been historically underrepresented in biomedical research. EHR data on more than 451,000 participants have been collected. For these participants, there is ongoing linkage to electronic health record (EHR) data, including ICD-9/ICD-10, SNOMED, and CPT codes. Genetic data comprises array samples from 315,000 participants and whole-genome sequencing (WGS) from 245,394 participants.

Ethics oversight

Informed consent for all participants is conducted in person or through an eConsent platform that includes primary consent, HIPAA Authorization for Research EHRs, and Consent for Return of Genomic Results. This research was approved by the Mass General Brigham institutional review board (protocol 2020P001737).

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Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	All above analyses were performed in 236,393 participants out of the 245,394 participants with WGS data available in All of Us. Of these 236,393 participants, 80% (n = 189,114) were used as the training cohort for PRSMix. Thus, all results detailed in the manuscript were calculated in the holdout testing cohort comprising the remaining 20% (n = 47,279) of participants.
Data exclusions	Participants missing basic demographic information (e.g. enrollment date, date of birth, sex assigned at birth) were excluded. Additionally, those with conflicting demographic information (e.g. sex assigned at birth differing from self reported gender) were excluded. This resulted in 236,393 participants remaining for analysis out of the 245,394 participants with WGS data available.
Replication	Careful steps were taken to obtain polygenic score weights developed using cohorts independent from All of Us. After training the PRSMix-based scores in a training subset of 189,114 All of Us participants, analyses of polygenic risk associations with coronary artery disease, type 2 diabetes mellitus, and major depressive disorder were validated in a separate cohort of 47,279 individuals.
Randomization	Not applicable as this is an observational study.
Blinding	Not applicable as this is an observational study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

Methods

n/a	Involved in the study	n/a	Involved in the study
☒	☐ Antibodies	☒	☐ ChIP-seq
☒	☐ Eukaryotic cell lines	☒	☐ Flow cytometry
☒	☐ Palaeontology and archaeology	☒	☐ MRI-based neuroimaging
☒	☐ Animals and other organisms		
☒	☐ Clinical data		
☒	☐ Dual use research of concern		
☒	☐ Plants		

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.